Differential Geometry Notes

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1 Preliminaries

Definition 1. *$N \subseteq M$ is a **submanifold** of M if N is a manifold.*

Definition 2. *A local parametrization $\phi : \Omega \to U$ of M, with inverse ψ is called compatible with S if $\psi(S \cap U) = \Omega \cap (\mathbb{R}^m \times \{0\})$*

Claim 1. *There is a local parametrization compatible with S*

Claim 2. *Suppose ϕ is compatible with S. define $W = \{x : (x, 0) \in \Omega\}$. then $\varphi : W \to S \cap U$ is a local parametrization of S defined by $x \to \phi(x, 0)$*

Definition 3. *The **tangent bundle** $TM \subseteq \mathbb{R}^{2k}$ of M is defined as $\{(p, v) : p \in M, v \in T_pM\}$*

Claim 3. *TM is a manifold of dimension $2n$*

Definition 4. *Suppose $\phi \in Diffeo(\Omega, U)$ is a local parametrization around p*

- *We define the radial map $r \in C^\infty(U)$ by $r(\phi(x)) = ||x||$*
- *If $\phi(0) = p$ then we define $\phi_r : U \setminus \{p\} \to \mathbb{R}^k$ by $\phi(a) \to (D\phi)_a \left(\frac{a}{||a||}\right)$*

2 Notions associated with the derivative

Definition 5. *Define $grad \in L(C^\infty(M), \mathcal{X}(M))$ by $\langle v, grad_f(p)\rangle = Df_p(v)$ for all $v \in T_pM$*

Definition 6. *Define $Hess_p \in L(C^\infty(M), T_pM^{(2)})$ by*

$$f \to ((v, w) \to \langle \nabla_v X_f, w\rangle)$$

Claim 4. *range $Hess_p \subseteq (T_pM)^{(2)}_{sym}$*

Claim 5. *The map $X, Y \to D_XY$ is bilinear and has the following properties:*

- *$D_{fX}Y = f(D_XY)$*

- $D_X(fY) = f(D_X Y) + (D_X f)Y$
- $D_X Y - D_Y X = [X, Y]$
- $D_X\langle Y, Z\rangle = \langle D_X Y, Z\rangle + \langle Y, D_X Z\rangle$

Claim 6. *Suppose $\phi : \Omega \to V$ is a local parametrization. Then $D_{\phi_i}\phi_j = \frac{\partial \phi}{\partial x_i \partial x_j}$*

3 Pullback by a diffeomorphism

Suppose $F : M \to N$ is a diffeomorphism

Claim 7. *Suppose $v \in T_p M$. Then for all $f \in C^\infty(M)$ we have $D_v(f \circ F) = (D_{F_* v})f$*

Claim 8. *We have $F_*[X, Y] = [F_* X, F_* Y]$*

Claim 9. *We have $\langle F_* X, F_* Y\rangle = \langle X, Y\rangle$*

4 Riemannian metric

Definition 7. *A diffeomorphism $F : M \to N$ is called a **Riemannian isometry** if at each point the derivative DF_p is an isometry from $T_p M$ to $T_p N$*

Definition 8. *A local diffeomorphism $F : M \to N$ at p is called a **local Riemannian isometry** at p if there is a neighborhood U such that $F(U)$ is open and $F_{|U} : U \to F(U)$ is a Riemannian isometry*

Definition 9. *Suppose $\gamma : [a, b] \to M$ is a piecewise smooth curve.*
*A **reparametrization** of γ is given by $\gamma \circ \phi$, where $\phi : [c, d] \to [a, b]$ is a homeomorphism with a partition $c_0, c_1, \ldots, c_k$ such that $\phi_{|[c_{i-1}, c_i]}$ is a diffeomorphism onto its image*

Claim 10. *For any γ piecewise regular, the map*

$$c \to \mathit{length}(\gamma_{[a,c]})$$

is a valid reparametrization, called the arc length parametrization

Suppose M is connected

Claim 11. *There is a piecewise regular curve between any two points*

Definition 10. *Define the **Riemannian metric** by*

$$d_M(p, q) = \mathit{inf}\{L(\gamma) : \gamma \text{ is a piecewise smooth curve from } p \text{ to } q\}$$

Claim 12. *The topology induced by the Riemannian metric is the same as the subspace topology M inherits from $\mathbb{R}^k$*

Definition 11. *A piecewise smooth curve γ from p to q is* ***length minimizing*** *if it has length $d_M(p,q)$*

Claim 13. *If there is a length minimizing curve from p to q then there is such a length minimizing piecewise regular curve as well*

Claim 14. *If M is closed in $\mathbb{R}^k$ then there is a length minimizing curve between any two points*

Claim 15. *Every Riemannian isometry is an isometry with respect to the Riemannian metric*

Claim 16. *Suppose F_1, F_2 are local Riemannian isometries with $F_1(p) = F_2(p)$ and $(DF_1)_p = (DF_2)_p$. Then $F_1 = F_2$*

5 First fundamental form

Definition 12. *The* ***first fundamental form*** *of M at p is defined by restricting the standard inner product on $\mathbb{R}^k$ to the tangent space. We have $g_p : T_pM \times T_pM \to \mathbb{R}$*

Definition 13. *For a local parametrization $\phi : \Omega \to V$ define*

$$G_ij(q) = g_q(\phi_i(q), \phi_j(q))$$

Claim 17. *Two manifolds M and N admit a local Riemannian isometry from p to q if and only if there are local parametrizations $\phi : \Omega \to U$ and $\varphi : \Omega \to V$ such that $G_{ij}(\phi(a)) = H_{ij}(\varphi(a))$ for all $a \in \Omega$*

6 Levi-Civita connection

Definition 14. *We have $\pi_p \in End(\mathbb{R}^k)$ which is the orthogonal projection onto T_pM. We define $\hat{\pi}_p$ as $I - \pi_p$*

Definition 15. *Suppose $v \in T_pM$. Define $\nabla_v \in L(\mathcal{X}(M), T_pM)$ by $X \to \pi_p(DX)_p v$*

Definition 16. *Suppose $X, Y \in \mathcal{X}(M)$ are smooth vector fields. Then we have the smooth vector field $\nabla_X Y = (p \to \pi_p(D_X Y)(p))$ called the* ***Levi-Civita connection*** *of X and Y*

Claim 18. *The map $X, Y \to \nabla_X Y$ is bilinear and has the following properties:*

- $\nabla_X Y - \nabla_Y X = [X, Y]$
- $\nabla_{fX} Y = f(\nabla_X Y)$
- $\nabla_X (fY) = f(\nabla_X Y) + (D_X f)Y$

Definition 17. *Suppose $\phi : \Omega \to V$ is a local parametrization. The **Christoffel symbols** $\Gamma_{ij}^k \in C^\infty(\Omega)$ are defined by $(\nabla_{\phi_i}\phi_j)(\phi(a)) = \sum_k \Gamma_{ij}^k(a)\phi_k(a)$*

Definition 18. *The **curvature tensor** of X, Y, Z is defined by $R(X,Y,Z) = \nabla_X(\nabla_Y Z) - \nabla_Y(\nabla_X Z) - \nabla_{[X,Y]}Z$*

Suppose $F : M \to N$ is a Riemannian isometry

Claim 19. *We have $\nabla_{F_*X}(F_*Y) = F_*(\nabla_X Y)$*

Claim 20. *We have $\nabla_{F_*v} \circ F_* = DF_p \circ \nabla_v$*

Claim 21. *We have $F_*[R(X,Y,Z)] = R(F_*X, F_*Y, F_*Z)$*

7 Vector fields along a curve

Definition 19. *Suppose $\gamma : I \to M$ is a smooth curve. A smooth vector field along γ is a map $V : I \to \mathbb{R}^k$ such that $V(t) \in T_{\gamma(t)}M$*

The vector space of smooth vector fields along γ is $\mathcal{X}(\gamma)$

Definition 20. *We have $cov \in L(\mathcal{X}(\gamma))$ given by $V \to (t \to \pi_{\gamma(t)}\dot{V}(t))$*

Definition 21. *A vector field V along γ is called **parallel** to γ if $V \in ker(cov_\gamma)$*

Claim 22. *Suppose $t_0 \in I$, $p_0 = \gamma(t_0)$ and $v_0 \in T_{p_0}M$. There is a unique parallel vector field V satisfying $V(t_0) = v_0$*

Definition 22. *Let $t_1, t_2 \in I$. We have the **parallel transport** map $P_{t_1t_2} \in L(T_{\gamma(t_1)}M, T_{\gamma(t_2)}M)$*

Claim 23. *$P_{t_1t_2}$ is an isometry*

Claim 24. *$P_{t_1t_2}$ is invertible with inverse $P_{t_2t_1}$*

Claim 25. *Suppose t_0 is in the interior of I. Then we have*

$$\nabla_v X = lim_{t \to t_0} \frac{P_{tt_0}(X(\gamma(t))) - X_p}{t - t_0}$$

8 Geodesics

Definition 23. *A smooth curve γ is a **geodesic** if γ' is parallel to γ*

Suppose $p \in M$ and $v \in T_pM$. Consider geodesics γ such that $\gamma(0) = p$ and $\gamma'(0) = v$

Claim 26. *There is such a geodesic γ on $(-\epsilon, \epsilon)$*

Claim 27. *Any two such geodesics agree on their common domain*

Claim 28. *There is a unique **maximal geodesic** γ_v of this kind on an open interval I_v*

Claim 29. *$I_{cv} = \frac{I_v}{c}$ and $\gamma_{cv}(t) = \gamma_v(ct)$*

8.1 Relation with length minimizing cuves

Claim 30. *If γ is a geodesic then for each t we have that $\gamma_{[a,b]}$ is length minimizing where $t \in [a,b]$*

Claim 31. *Suppose γ is piecewise regular and length minimizing. Then γ is smooth*

Claim 32. *Suppose γ is regular and length minimizing. Then the arc length parametrization of γ is a geodesic*

9 Exponential map

Definition 24. *We define $E_p = \{v \in T_pM : 1 \in I_v\}$*

Claim 33. *E_p is star convex with respect to 0*

Claim 34. *E_p is open in T_pM*

Claim 35. *$I_v = \{t : tv \in E_p\}$*

Definition 25. *We define $\exp_p \in C^\infty(E_p, M)$ by $v \to \gamma_v(1)$*

Claim 36. *$(D\exp_p)_0 = I$*

Claim 37. *$\exp_p$ is a local diffeomorphism at 0*

Claim 38. *$\gamma_v(t) = \exp(tv)$*

Definition 26. *We define $E \subseteq TM$ as $\{(p,v) : v \in E_p\}$*

Claim 39. *E is open in TM*

Definition 27. *We define $\exp \in C^\infty(E, M)$ by $(p,v) \to \exp_p(v)$*

Suppose $F : M \to N$ is a local Riemannian isometry at p

Claim 40. *For all v in T_pM we have $F_*(v) \in \mathcal{E}_{F(p)}$*

Claim 41. *We have $\exp_{F(p)}(F_*(v)) = F(\exp_p(v))$*

10 Normal coordinate systems

Definition 28. *A neighborhood $U \subseteq E_p$ of the origin is called **prenormal** if $\exp_p : U \to \exp(U)$ is a diffeomorphism.*

Definition 29. *A **normal neighborhood** is the image under $\exp_p$ of a star convex prenormal neighborhood*

Definition 30. *Suppose $v_1, v_2, \ldots, v_n$ is an orthonormal basis of T_pM and U is a normal neighborhood of p with corresponding prenormal neighborhood V. Define $\Omega = \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i v_i \in V\}$ and $\phi \in Diffeo(\Omega, U)$ by $\phi(x) = \exp_p(\sum x_i v_i)$ Then ϕ is called a **normal coordinate system** centered at p*

Claim 42. *If ϕ is a normal coordinate system on U then $\phi_i(p)$ is an orthonormal basis of T_pM*

Claim 43. *If $\phi \in Diffeo(\Omega_1, U)$ and $\psi \in Diffeo(\Omega_2, U)$ are normal coordinate systems then there is $A \in O_n$ such that*

- $\Omega_2 = A(\Omega_1)$
- $\psi \circ A \circ \phi^{-1} = id_U$

Claim 44. *Suppose $\phi \in Diffeo(\Omega, U)$ is a normal coordinate system centered at p with $\phi_i(p) = v_i$. Then the following hold:*

- $\Gamma_{ij}^k(0) = 0$
- $G_{ij}(p) = \delta_{ij}$
- *Whenever $tx \in \Omega$ we have $\gamma_v(t) = \phi(tx)$ where $v = \sum x_i v_i$*
- $(DG_{ij})_p(D\phi)_0 = 0$
- *For $q = \exp_p(v) \in U$ we have*
  - $r(q) = ||v||$
  - $||v||\phi_r(q) = \gamma_v'(1)$
  - $||\phi_r(q)|| = 1$
- *grad $r = \phi_r$*

11 Injectivity radius

Definition 31. *The **injectivity radius** is defined as*

$$inj_p = sup\ \{r > 0 : B_r \text{ is prenormal}\}$$

Claim 45. *If $\rho = inj_p$, then B_ρ is prenormal*

Claim 46. *Suppose $0 < \epsilon < \rho$. Then $\Sigma_\epsilon = \exp_p(\delta B_\epsilon)$ is a submanifold of M*

Claim 47. *Suppose $\phi \in Diffeo(\Omega, U)$ is a normal coordinate system centered at p. Let $q = \exp_p(v) \in U$ with $\epsilon = ||v|| < \rho$. Then $\phi_r(q) \in T_q\Sigma_\epsilon$*

Claim 48. *Suppose $q = \exp_p(v)$ where $||v|| < \rho$. Consider the curve $\gamma : [0, 1] \to M$ given by $\gamma(t) = \exp_p(tv)$. Then we have the following:*

- *γ is minimizing*
- *Any other minimizing curve from p to q is a reparametrization of γ*

Definition 32. *A neighborhood V of p is δ- uniformly normal if for all $q \in V$ we have*

- $\delta < inj_q$
- $V \subseteq \exp_q(B_\delta)$

Definition 33. *A neighborhood is uniformly normal if it is δ-uniformly normal for some δ*

Claim 49. *There is a uniformly normal neighborhood around every point*

12 One parameter family of curves

Definition 34. *We call $\Gamma \in C^\infty(I \times J, M)$ a **one parameter family of curves** on J*

Definition 35. *The **variation field** $V \in \mathcal{X}(\Gamma_0)$ of Γ is defined by $t \to \partial_1\Gamma(0,t)$*

Claim 50. *We have $[cov(\partial_1\Gamma)_s](t) = [cov(\partial_2\Gamma)_t](s)$*

Suppose $\gamma : [a,b] \to M$ denotes a regular curve from p to q

Definition 36. *A **proper variation** of γ is a map $\Gamma \in C^\infty((-\epsilon,\epsilon) \times K, M)$ such that $\Gamma_0 = \gamma$ and Γ_s is a regular curve from p to q*

Claim 51. *Suppose $V \in \mathcal{X}(\gamma)$ with $V(a) = V(b) = 0$. Then there is a proper variation Γ of γ whose variation field is V*

13 Second fundamental form

Definition 37. *N_pM is the orthogonal complement of T_pM in $\mathbb{R}^k$*

Definition 38. *The **second fundamental form** of M at p is the bilinear map $S_p : T_pM \times T_pM \to \mathbb{R}^k$ given by $v, w \to [(D\pi)_p v]w$*

Claim 52. *range $S_p \subseteq N_pM$ for all p*

Claim 53. *The second fundamental form is symmetric*

Claim 54. *We have $S_p(X_p, Y_p) = \hat{\pi}_p((D_X Y)(p))$*